

Vital Sleep Patch:

Transforming Pediatric Sleep Apnea Diagnosis through Home-Based Testing

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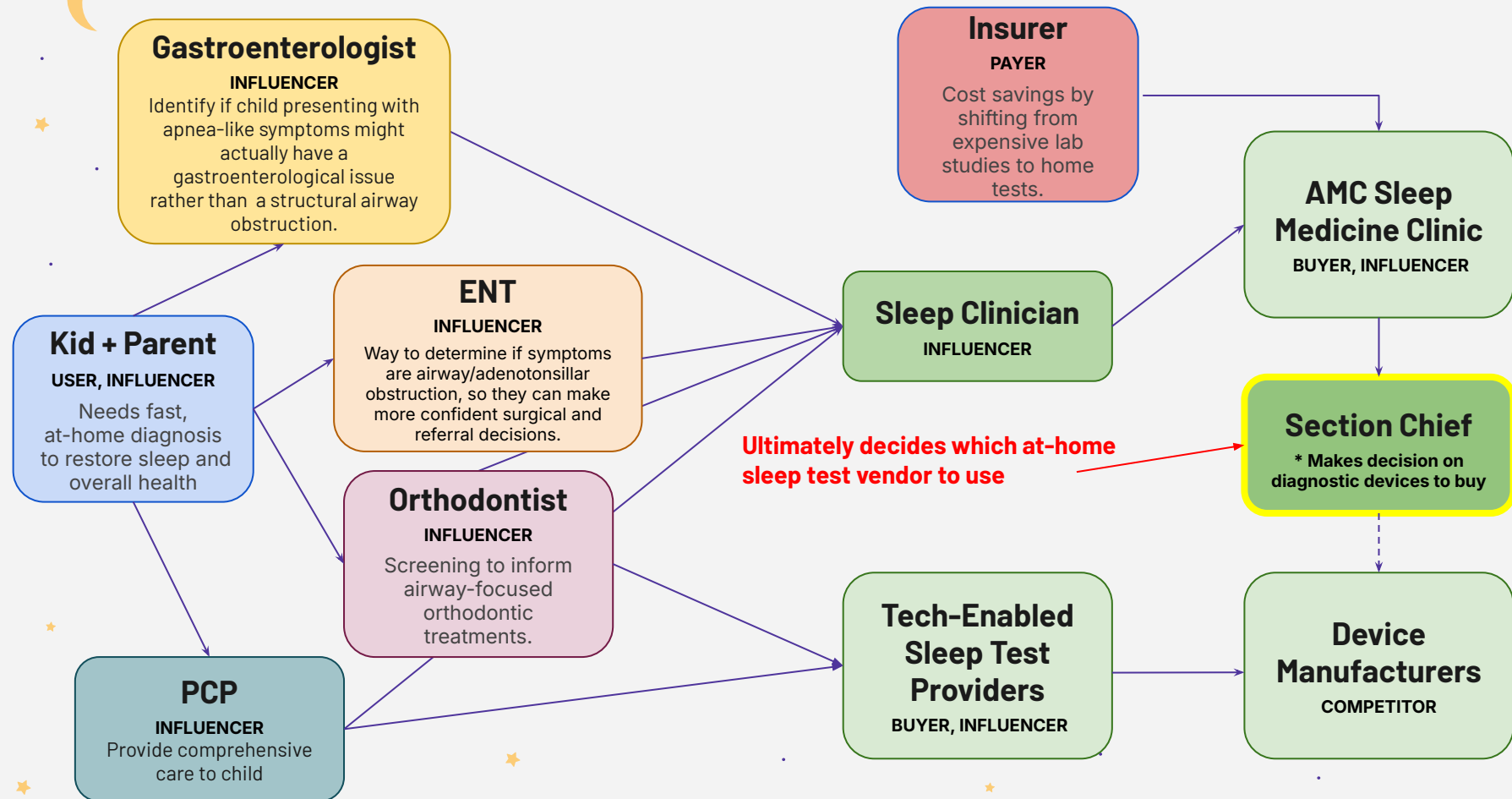
Week 3 Interviews Conducted: 9 human + 5 synthetic

Total Interviews: 29 human + 10 synthetic

★ **Problem Solving Journey:** Started with a wearable sensor patch and AI-based diagnostic pipeline for pediatric OSA. Now, through customer discovery, we are focusing on understanding our customer's workflow and desired outcomes.



Updated Customer Pathway and Decision Makers



Day in the Life of an AMC Sleep Medicine Clinician

Insurance
requirement

Adults

- 01 Symptoms / Referral:** Patients present with snoring, witnessed apneas, daytime sleepiness, obesity, or other risk factors
- 02 Evaluation:** Clinical question: can this patient start with a home test, or do they need full in-lab PSG? 80% begin with HSAT
- 03 Home Sleep Test:** Captures limited respiratory signals; useful as a first-pass test
- 04 Scoring:** Results may be diagnostic or non-diagnostic. Negative HSAT does not rule out OSA
- 05 Escalation:** If HSAT is non-diagnostic, technically limited, or clinician determines for another reason that it was insufficient, proceed to in-lab PSG
- 06 Diagnosis + Treatment:** Confirm severity. Prescribe CPAP

Section Chief of Sleep Medicine decides which device to buy

Pediatrics

- 01 Referral :** Child has snoring, breathing pauses, restless sleep, behavioral concerns, obesity, or ENT-related concerns; referred by PCP or another specialty.
- 02 Evaluation:** Different diagnostic thresholds; subtle disease remains clinically important
- 03 In-lab PSG:** In-lab PSG is usually required; home testing would be valuable, but current systems are limited
- 04 Arrival:** Arrive 6:30–7:00 PM. Caregiver will stay overnight in lab.
- 05 Hookup and Acq:** 25–35 wires; setup takes ~1 hour. Active overnight monitoring. Techs actively troubleshoot (usually each tech will have 2 patients). Parents stay overnight
- 06 Scoring:** Manual diagnosis. Goal is not a “perfect” study, but enough reliable physiologic data
- 07 Diagnosis + Treatment:** Determine clinical severity and primary intervention (e.g., T&A or PAP)

Pain
Points

Operational Pain Points for AMC Sleep Medicine Clinicians

Operational Pain Points

- 25–35 wires; ~1 hour setup
- Kids move / remove sensors; ~1 in 50–60 studies may fail
- Critical signals are hard to maintain: nasal flow, TC02, effort belts, SpO2, EKG
- Techs actively troubleshoot all night; ~2 patients/tech
- Caregiver overnight required; parent distress is a major barrier
- Special-needs children increase complexity
- Workflow mismatch: arrival 6:30–7 PM, hookup 7:30–8 PM, limited flexibility for early sleepers
- Capacity constraints: ~10 pediatric studies/week, booked into September, tertiary waits ~1 year

Unmet needs we can help solve

- Wireless / untethered monitoring
- Fewer sensors, less intimidating setup
- Robust to movement and real-world child behavior
- Enough data for diagnosis, not a perfect study
- Reduce backlog and improve access

Takeaway: The core pain point is getting reliable diagnostic data in a child-friendly at-home form factor, and in an operationally feasible way.

Day in the Life/Journey: Child with OSA

Clinical

- 01 PCP Visit:** Patient presents with obesity. Inquiry reveals nighttime snoring, daytime sleepiness / decreased test scores / increased irritability.
- 02 Referral:** PCP may refer to sleep specialist (AMC) or ENT (rural). Otherwise, pt may get dental attention
- 03 Dentist/Orthodontist:** Captures oral causes or complications (crowded lower jaw, swollen/inflamed tonsils). Recommends OSA test for dental appliance.
- 04 Testing:** May refer to an online testing provider, local testing facility with pediatric accreditation, or AMC.
- 05 Diagnosis + referral:** AMC test determines moderate or severe sleep apnea. Sleep doctor recommends surgery
- 06 Treatment:** ENT performs T&A (tonsillectomy+/adenoidectomy) and evaluates need for CPAP use

Home

- 01 Referral :** Child has snoring, breathing pauses, restless sleep, behavioral concerns, obesity, or ENT-related concerns
- 02 Evaluation:** Parent may schedule ad-hoc or surface at scheduled annual PCP visit
- 03 Assessment:** PCP or dentist recommends OSA evaluation, parent decides to follow/choose tester
- 04 Testing:** Decides to go to AMC. Wait 6 months. Have to arrive 5:30–7:00 PM. Caregiver will stay overnight in lab.
- 05 Hookup and Acq:** 25–35 wires; setup takes ~1 hour. Active overnight monitoring. Techs actively troubleshoot (usually each tech will have 2 patients). Parents stay overnight
- 06 Scoring:** Wait another 2–3 months. Get result recommending surgery/CPAP/monitoring
- 07 Treatment + Monitoring:** Support surgical recovery (stay at home, food), ensure CPAP compliance over months

Patient Journey: from Months to Days



Maya is an 11-year-old high-risk (obese) patient whose PCP has referred her to an ENT for evaluation of snoring and disordered breathing.

Current Journey

ENT schedules PSG

4-6 month waitlist →

Overnight PSG in clinic

3-4 month backlog →

Scorer scores study

→

Physician confirms diagnosis

→

ENT schedules surgery

with Vital Sleep Patch

ENT provides patch

same day →

Sleep at home with patch

3 days →

AI scores study

→

Physician confirms diagnosis

→

ENT schedules surgery

	Now	Vital Sleep Patch	Benefit?
Time to Diagnosis	9 months	1 week	30x faster
Cost of Diagnosis	\$1k OOP	\$300 OOP	70% cheaper

Synthetic Interviews

SEGMENT 1 · Pediatric GI Specialist

Dr. Maya Patel

- ✓ OSA overlap is common (~30–40%) and drives diagnostic uncertainty
- ✗ Assumption that a cash-pay D2C model works here → Heavy Medicaid populations at AMCs cannot access \$300 OOP tests.

"I need a binary 'high risk' flag to bypass the 6-month lab wait and send them straight to the ENT surgeon."

SEGMENT 2 · Orthodontist

Dr. James Chen

- ✓ Clinics will bypass CPT codes entirely by bundling the device cost into high-margin airway interventions.
- ✗ Assumption that referrals convert → most patients never complete them

"Telling parents to go see another doctor kills momentum."

SEGMENT 3 · Sleep Lab

Dr. Robert Langston

- ✓ Pediatric PSGs are hospital loss-leaders; labs want to offload mild cases to save beds for complex interventions
- ✗ Assumption that it could replace full sleep studies → not viable for complex cases

"This helps us triage—but it's not a replacement."

SEGMENT 4 · OBESITY CLINIC

Dr. Lisa Fernandez

- ✓ OSA diagnosis is required to unlock treatment approval
- ✗ Assumption that deep respiratory data is valued → Clinics strictly need a simple, "insurance-legible" PDF to clear prior-auths.

"If this speeds up approval, it changes everything."

SEGMENT 5 · Chief Medical Officer

Dr. Sarah Okafor

- ✓ Adoption depends on proven cost savings and outcomes
- ✗ Assumption that clinical accuracy alone drives adoption → financial proof is required

"Show me cost savings—not just that it works."

Hypotheses [1/2]



PARENTS

H1 Doctor endorsement is the primary trust signal



David: In pediatrics, when doctor says 'this is what I recommend for your child,' decision effectively made

H2 Parents pay \$200–\$400 OOP if clinically validated



Tyler Ogren confirmed ~\$200–300 range; Empower uses OOP superbills.

H3 Reimbursement as a trust signal



Aditya: Reimbursement is a legitimacy and trust signal, independent of affordability.

H4 Form factor comfort is a hard requirement



Empower: Form factor is a top reason for us to sign up new partners

H5 Mail-in patch model is acceptable if workflow is clean



Empower Sleep uses mail-in model with return label — works in practice.

H6 Rural families have higher hurdle tolerance



INDEPENDENT ENTs / PCPs / ORTHO

H1 In-house diagnostics meaningfully improve surgical conversion



Empower Sleep: dental/ortho act as massive B2B channels; use as pre-intervention diligence.

H2 Workflow simplicity beats clinical perfection



Curt Langlotz: piggyback existing data systems; Empower requires open APIs.

H3 CPT code is a hard prerequisite for routine use



David Kopp + Empower: can scale B2B via cash-pay/superbills before formal CPT.

H4 Physicians won't 'sell' OOP healthcare to patients



H5 Dental/Ortho is a faster-converting channel than PCPs



NEW: Tyler Onderdonk — dentists are largest untapped B2B referral channel.



OBESITY / GLP-1 CLINICS

H1 OSA screening is a hard GLP-1 insurance prerequisite



Aditya: GLP-1 in pediatrics does not seem constrained by insurance
David: Deprioritize GLP-1s until you get evidence

H2 Documentation compliance > clinical precision



Validated



Invalidated



New Hyp.



Not Discussed in W2

Hypotheses [2/2]



PAYORS

H1 Payors need published health-economic case before opening pathways



Not yet discussed with payor contacts.

H2 Value-based care payors more motivated than FFS payors



Not yet discussed with payor contacts.



HOSPITALS & AMCs

H1 AMCs accept startups only as 'triage' tools, not competitors



Validated by Aditya

H2 Two most important physiologic channels for diagnosing pediatric OSA are: Airflow, Respiratory effort



Initial indication by Dr. Greenbough

H3 Clinical adoption requires physician reviewable raw data, not just an automated summary score.



Dr. Greenbough confirmed



TECH-ENABLED PROVIDERS & INCUMBENTS

H1 Partner with adult full-stack providers as pediatric hardware arm



NEW: Tyler Onderdonk — scale by white-labeling to existing adult IDTF/providers vs. building from scratch.

H2 License hardware/algorithms to dominant adult players



NEW: David Kopp — faster, more capital-efficient than building full-stack D2C pediatric clinic.

H3 Incumbents will license our tech for capabilities they lack



NEW: David Kopp — positional data + central vs. obstructive differentiation are major clinician differentiators.



Other target customers (deprioritized)

H1 Military/VA will adopt screening at enlistment to prevent 'pre-existing' sleep apnea disability payouts

H2: Multi-child households have more logistical strain and might be willing to invest more



Validated



Invalidated



+ New Hyp.



— Not Discussed in W2



Experiments

Testing MVP and hypotheses against market reality: Testing whether our current prototype delivers the exact sensor data and workflow simplicity required to trigger adoption.

Conferences to learn more:

- Considering attending the American Academy of Sleep Medicine annual meeting in Baltimore (May 3rd) and/or the California Sleep Society meeting in Sonoma (September 17-19)

How we are testing (interviews / showcasing physical MVP / working on digital MVP):

- Conducting competitive analysis against existing pediatric and adult sleep testing solutions
- Comparing our MVP on: 1) Pediatric-first design 2) Multi-modal clinical sensing 3) Integrated sensing in a single device 4) AI-enabled diagnostic workflow 5) Validated in pediatric populations
- Pitching our feature set directly to independent clinics, payors, and adult market proxies (e.g., Empower Sleep) to separate the "must-haves" from the "nice-to-haves"

What the experiments validate:

- Are we a pre-PSG screening tool or a full diagnostic replacement?
- Where do we definitively outperform the competition, and what hardware gaps must we close?
- Does this MVP make us a full-stack service provider, or an OEM licensing to incumbents?

Experiment / MVP #1 - Validating financials

Objective & Model Logic

Determine whether adding a pediatric at-home test pathway can improve clinic economics by expanding access and relieving bottlenecks.

Main Questions:

Does a pediatric home-testing pathway create more total value by increasing completed diagnoses and preserving in-lab capacity?

How does our at-home system fit in with the financial incentives of the AMC sleep clinic?

Patient Flow:

Referral → Consult → Home Test / PSG →
Diagnosis → Treatment

What the MVP model will include:

- Revenue and cost per consult, home test, lab PSG, and downstream follow-up
- Capacity effects: home testing freeing scarce lab openings for high-complexity cases
- Full cycle flow from referral to diagnosis and treatment

Minimal Model Outputs:

- Contribution margin per new patient
- Incremental value of freeing one lab PSG slot

Hypothesis: Value is created by incorporating an at-home sleep testing device in the pediatric cohort.

Access optimization – diagnose more children without adding sleep-lab capacity

Triage optimization – send lower-complexity patients home first and preserve PSG for the highest-need cases

Throughput optimization – increase completed diagnostic workups across the clinic

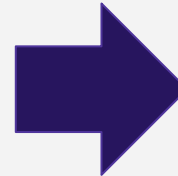
Experiment / MVP #2 - Orientation of sleeping

Feedback from interviews:

Customers want a specific analysis regarding **positional orientation**.

David Kopp mentioned this in multiple calls:

*"Positional data is clinically useful for patient-centric recommendations. Example:
"You have a lot of apneas when you're sleeping on your left side; you should sleep on your right side because it goes away."*



Iterating on the prototype / MVP:

Updating data analysis algorithms to monitor and interpret positional data

Showcase new dashboard and solicit feedback from testers + doctors



Pivot/Proceed Checkpoints

Proceed if:

- Interviews identify that our MVP solves a real unmet need and stakeholders identify a credible adoption pathway
- Our current sensor set is sufficient for the intended initial use case
- We validate a clear path to early monetization, either via out-of-pocket superbills or a direct B2B licensing partnership
- The product can fit into an existing clinical or commercial workflow without major friction

Pivot if:

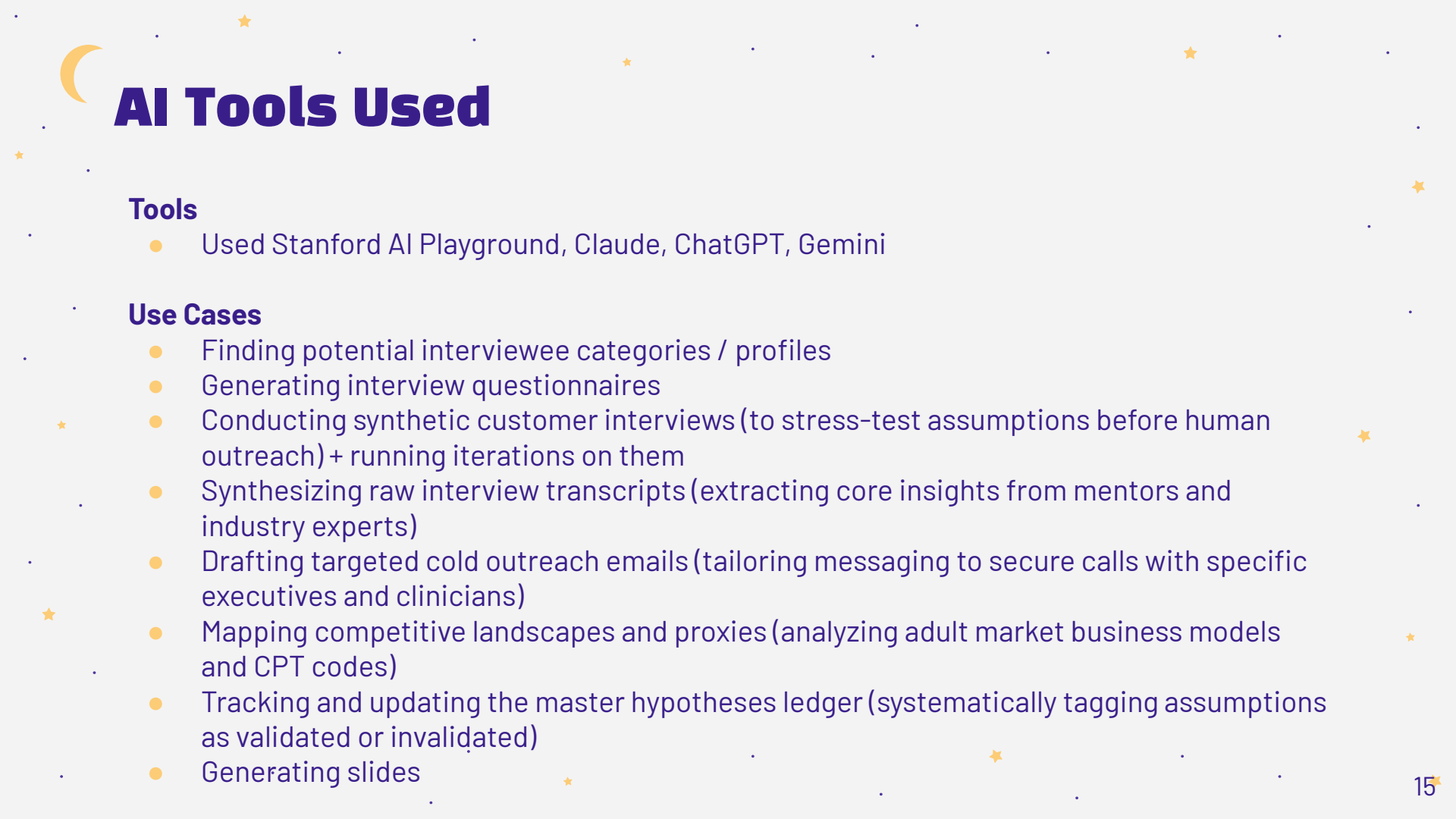
- Key stakeholders indicate that our solution is not important or valuable enough to warrant adoption
- Important sensors or features are missing for adoption
- Our differentiation versus competitors is not clear
- Buyers do not see enough value
- Creates added workflow burden that is too high for clinics, intermediaries, or families
- Becomes clear that the best initial use case is different from our current target

To-do Next Week and Help needed!

- Validate **Regulatory and Reimbursement Pathway**: Clear answer to "Will insurance cover this?" — identified as #1 adoption barrier across ALL stakeholder interviews
- Focus on **Academic Medical Centers (AMCs)** as Initial Channel:
 - Conduct interviews with AMC sleep lab directors, pediatric specialists (GI, pulmonology, ENT), and hospital administrators to validate triage/throughput value proposition
 - Address key AMC concerns: co-validation against PSG, clear clinical guidelines for when to use Vital vs. full sleep study, integration with existing referral workflows
- More interviews and **digital MVP** (have concept in place now)

Help needed

- Intros to clinicians to test our hypotheses around providers, esp. across different geographies and socio-demographics
- Intros to value-based care organizations, regulatory experts, and payors



AI Tools Used

Tools

- Used Stanford AI Playground, Claude, ChatGPT, Gemini

Use Cases

- Finding potential interviewee categories / profiles
- Generating interview questionnaires
- Conducting synthetic customer interviews (to stress-test assumptions before human outreach) + running iterations on them
- Synthesizing raw interview transcripts (extracting core insights from mentors and industry experts)
- Drafting targeted cold outreach emails (tailoring messaging to secure calls with specific executives and clinicians)
- Mapping competitive landscapes and proxies (analyzing adult market business models and CPT codes)
- Tracking and updating the master hypotheses ledger (systematically tagging assumptions as validated or invalidated)
- Generating slides